

PRASAD'S PHYSICS

BRAHMAPUR

Assignment

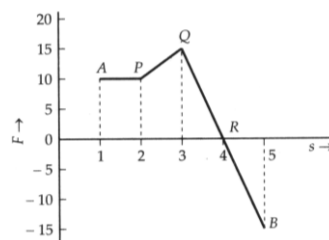
Work, Energy, and Power

Numerical Problems

01. A box is pushed through 4.0m across a floor offering 100 N resistance. How much work is done by the (i) applied force (ii) resisting force?
02. A force $F = (10 + 0.5x)$ acts on a particle in x direction. Find the work done by this force during a displacement from $x = 0$ to $x = 2m$.
03. If the linear momentum of a body is increased by 10%. What is the % change in its kinetic energy?
04. A ball is dropped from a height h_0 on to a horizontal plane. Find the height to which rebounds after collision if e is the coefficient of restitution of the collision.
05. A rubber ball falls on a floor from a height of 19.6 m. Calculate the velocity with which it strikes the ground. To what height will the ball rebound if it loses 25% of its energy on striking the ground?
06. A body of mass 2 kg makes an elastic collision with another body at rest and continues to move the original direction with a speed equal to one third of its original speed. Find the mass of the second body.
07. A body of mass 5kg initially at rest is subject to a force of 20 N. What is the kinetic energy acquired by the body at the end of 10s?
08. A railway carriage of mass 10000kg moving with a speed of 15 ms^{-1} strikes a stationary carriage of the same mass. After the collision, the carriages get coupled and move together. What is their common speed after collision?
09. A bullet of mass 50g moving with a velocity 400 m/s strikes a wall and goes out from the other side with a velocity of 100 m/s. Find work in passing through wall?
10. An engine of power 1.5 MW applies a force for 6 min. on a train moving with a velocity of 10 m/s and attains a velocity of 25 m/s. Neglecting friction, find mass of train?
11. A 2 kg mass moving with a velocity of 10 m/s collide with another 6kg mass moving in opposite direction

with a velocity of 20 m/s and stick together. Find their common velocity and momentum.

12. A ball strikes the floor vertically and rebounds with $e = 0.6$. If the mass of the ball is 0.01 kg and its velocity is 5 m/s just before it hits the ground. What is the velocity and momentum of the ball as it leaves the floor?
13. A man lifts 100 stones to a height of 6m in 2 min. If mass of each stone is 1 kg calculate the average power. [$g = 10 \text{ m/s}^2$]
14. A body moves from point A to B under the action of a force, varying in magnitude as shown in fig. Obtain the work done. Force is expressed in Newton and displacement in metre.



15. A force $\vec{F} = \hat{i} + 5\hat{j} + 7\hat{k}$ acts on a particle and displaces it through $\vec{s} = 6\hat{i} + 9\hat{k}$. Calculate the work done if the force is Newton and displacement in metre.

Multiple Choice Questions

01. The work done by the force $\vec{F} = (-\hat{i} + 2\hat{j} + 3\hat{k}) \text{ N}$ in moving the body a distance of 4m along z-axis is :
(a) 6 J (b) 18 J (c) 2 J (d) 12 J
02. When a truck moving with velocity v collides against a stationary drum, then drum will move with:
(a) same velocity of truck
(b) twice the velocity of truck
(c) half the velocity of the truck
(d) thrice the velocity of truck
03. An engine develops 10k W of power. How much time will it take to lift a mass of 200 kg to a height of 40 m ($g = 10 \text{ ms}^{-2}$)
(a) 4s (b) 5s (c) 8s (d) 10s

04. A body of mass 5kg initially at rest is subject to force of 20N. The K.E. acquired by the body at the end of 10 sec is :
(a) 4000 J (b) 400 J (c) 40 J (d) 4 J
05. In the stable equilibrium position, a body has :
(a) Maximum P.E. (b) Minimum P.E.
(c) Minimum K.E (d) Maximum K.E.
06. A body falls from height of 16 m rebounds to a height of 4m. The coefficient of restitution is :
(a) 1/4 (b) 1/2 (c) 3/4 (d) 1/8
07. If K.E. of a body is equal to it's momentum then it's velocity in m/s is :
(a) 1 (b) 2 (c) 3 (d) 4
08. The coefficient of restitution of collision has the range :
(a) $0 < e < 1$ (b) $0 \leq e \leq 1$
(c) $-1 \leq e \leq 1$ (d) None.
09. Mud thrown on the wall & sticking to it is an example of :
(a) perfectly elastic collision
(b) perfectly elastic collision
(c) inelastic collision
(d) perfectly inelastic collision
10. Work is done only when :
(a) applied force is strong
(b) applied force generates motion
(c) applied force is normal to the displacement
(d) force is applied
11. When the bob of a simple pendulum swings, the work done by tension in the string is :
(a) >0 (b) <0 (c) zero (d) maximum
12. For an elastic one-dimensional collision if the relative velocity of approach is v then the relative velocity of separation will be
(a) v (b) $-v$ (c) zero (d) $2v$
13. For all types of collisions; which one is / are conserved?
(a) momentum (b) K.E.
(c) total energy (d) both a & c
14. If the amount of work done by a force depends only on the initial and final positions of the object which has been moved, then such a force is called?
(a) gravitational (b) dissipative
(c) conservative (d) retarding
15. A particle of mass m moving with velocity v strikes a stationary particle of mass $2m$ and sticks to it. The speed of the system will be :
(a) $v/2$ (b) $2v$ (c) $v/3$ (d) $3v$.
16. A long spring is stretched by 2 cm. Its potential energy is V . If the spring is stretched by 10 cm, its potential energy would be :
(a) $V/25$ (b) $V/5$ (c) $5V$ (d) $25V$
17. A billiard ball moving with a speed of 5 m/s collides with an identical ball, originally at rest. If first ball stops dead after collision, then the second ball will move forward with a speed of :
(a) 10 m/s (b) 5 m/s
(c) 2.5 m/s (d) 1.0 m/s
18. A bullet in motion hits and gets embedded in a solid block resting on a frictionless table. What is conserved?
(a) momentum and kinetic energy
(b) momentum alone
(c) kinetic energy alone
(d) neither momentum nor kinetic energy
19. A ball is dropped on the ground from a height of 1m. The coefficient of restitution is 0.6. The height to which the ball will rebound is :
(a) 0.6 m (b) 0.4 m
(c) 0.36 m (d) 0.16 m
20. A rod of mass m and length ℓ is lying on a horizontal table. Work done in making it stand on one end will be :
(a) $mg \ell$ (b) $mg \ell / 2$
(c) $mg \ell / 4$ (d) $2 mg \ell$